

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	1 / 10

1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

Product information

Trade name MetAMINO®
DL-Methionine, Feed Grade 99%

Company Evonik Industries AG
Health & Nutrition
Abteilung HN-M-QR
Postfach 1345
D-63403 Hanau

Telephone +49 (0)6181/59-2024
Telefax +49 (0)6181/59-72024
Email address FA-MSDS@evonik.com
Emergency telephone number +49 (0)2236/76-2222
Emergency telephone number(Telefax) +49 (0)2236/76-2026

Use of the Substance / Preparation Feed Additive

2. HAZARDS IDENTIFICATION

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Not a hazardous material according to the Directive (EEC) No. 1272/2008 .

GHS-Labeling

Statutory basis Labelling not required according to EU-CLP Ordinance (1272/2008).

Classification as per Directive 67/548/EC or Directive 1999/45/EC

In accordance with EC directives or respective national laws, the product does not need to be classified nor labelled.

Other Hazards

Not applicable.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Information on ingredients / Hazardous components As per EU-CLP Ordinance (EC) No. 1272/2008

• DL-Methionine		>= 99%	
CAS-No.	59-51-8	EC-No.	200-432-1
Remarks	Labelling not required according to EU-CLP Ordinance (1272/2008).		

Information on ingredients / Hazardous components as per Directive 67/548/EC or Directive 1999/45/EC

• DL-Methionine		>= 99%	
CAS-No.	59-51-8	EC-No.	200-432-1

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	2 / 10

In accordance with EC directives or respective national laws, the product does not need to be classified nor labelled.

Texts of H phrases, see in Chapter 16
See chapter 16 for text of risk phrases

4. FIRST AID MEASURES

Inhalation

In case product dust is released:
Possible discomfort: cough, sneezing
Move victims into fresh air.

Skin contact

No hazards which require special first aid measures.

Eye contact

Possible discomfort is due to foreign substance effect.
Rinse thoroughly with plenty of water keeping eyelid open.
In case of persistent discomfort: Consult an ophthalmologist.

Ingestion

Have the mouth rinsed with water.
After absorbing large amounts of substance:
Consult a physician.

Most important symptoms and effects, both acute and delayed

Indication of any immediate medical attention and special treatment needed

After absorbing large amounts of substance:
Possible discomfort: nausea, vomiting
Treatment of symptoms, administration of activated charcoal, acceleration of the gastro-intestinal tract.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media

water, foam, mist

Unsuitable extinguishing media

carbon dioxide (CO₂)

Special hazards arising from the substance or mixture

May be released in case of fire: hydrocyanic acid, flammable smouldering gases, NO_x.
sulphur oxides, carbon monoxide, carbon dioxide.

Special protective equipment for fire-fighters

In the event of fire, wear self-contained breathing apparatus.

Advice for firefighters

Contaminated fire-extinguishing water must be disposed of in accordance with the regulations issued by the appropriate local authorities.
Fire residues should be disposed of in accordance with the regulations.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Wear personal protective equipment.
Keep unauthorized persons away.

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	3 / 10

Environmental precautions

Do not allow the product into the following compartments:
groundwater

Methods and material for containment and cleaning up

Absorb mechanically avoiding production of dust.

7. HANDLING AND STORAGE

Handling

Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practices.

Advice on protection against fire and explosion

Combustible

Keep away from sources of ignition - No smoking.

Avoid dust formation.

Must be rendered inert for grinding.

VDI 2263 "Dust fires and dust explosions; dangers, evaluation, preventive measures."

Ensure there are sufficient retaining facilities for water used to extinguish fire.

Dust explosion class

St1

Method: VDI 2263
Standardized max. rate of pressure increase, KSt: 85 bar·m/s

1 m3 standard container, ignition energy 10 kJ

Storage

Conditions for safe storage, including any incompatibilities

Keep in a dry, cool place.

Avoid light effect.

Keep container tightly closed and dry.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Engineering measures

Ensure suitable suction/aeration at the work place and with operational machinery.

Take precautionary measures against static discharges. Earthing of equipment.

Personal protective equipment

Respiratory protection

No special protective equipment required.

If dust occurs: dust mask with P1 particle filter

Hand protection

Glove material Nitrile, for example, Dermatril 740, Kächele-Cama Latex GmbH (KCL), Germany

Material thickness 0,11 mm

Break through time 8 h

Method DIN EN 374

Glove material Natural rubber (NR), for example, Cama Clean 708, Kächele-Cama Latex GmbH (KCL), Germany

Material thickness 0,5 mm

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	4 / 10

Break through time 8 h
Method DIN EN 374

Eye protection

Safety glasses with side-shields
If dust occurs: basket-shaped glasses

Skin and body protection

No special protective equipment required.

Hygiene measures

Wash face and/or hands before break and end of work.
Cleanse and apply cream to skin after work.

Protective measures

Handle in accordance with good industrial hygiene and safety practices.
If there is the possibility of skin/eye contact, the indicated hand/eye/body protection should be used.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance

Form crystalline
Colour colourless - light brown
Odour characteristic
Smell threshold: ca.3 ppm
(literature value)

Information on basic physical and chemical properties

pH 5 - 6 (10 g/l) (20 °C)
water

Melting point/range 266 - 267 °C
decomposition

Boiling point/range not applicable

Flash point not applicable

carbonisation point 210 °C

Evaporation rate No data available

Flammability (solid, gas) Combustible solid.
Method: VDI 2263

Ignition temperature 330 °C
Method: VDI Guideline 2263 sheet 1
(BAM-furnace)
Standard commercial product with characteristic grain size distribution is normally flammable.

glow temperature > 400 °C
Method: VDI 2263

Autoinflammability > 150 °C
Method: VDI Guideline 2263 sheet 1
in 8l wire basket

SAFETY DATA SHEET (EC 1907/2006)**MetAMINO®****DL-Methionine, Feed Grade 99%**

Material no.	Version	5.5 / REG_EU
Specification	Revision date	22.11.2011
VA-Nr	Print Date	28.11.2011
	Page	5 / 10

Thermal decomposition	215 °C TG (thermal gravimetric analysis)
Oxidizing properties	Not to be expected in view of the structure
Explosiveness	Not to be expected in view of the structure
Lower explosion limit	dust: 30 g/m ³
Upper explosion limit	No data available
Minimum ignition energy	10 - 30 mJ (25 °C) Method: VDI Guideline 2263 sheet 1 mean grain size: < 50 µm grain size < 63µm Normal combustability with inductance 30 - 100 mJ (25 °C) Method: VDI Guideline 2263 sheet 1 mean grain size: < 50 µm grain size < 63µm Normal combustability without inductance
maximum absolute explosive pressure	7,8 bar
Vapour pressure	< 0,0000001 hPa Method: calculated Modified Grain Method
Density	1,34 g/cm ³
Relative density	No data available
Bulk density	610 - 750 kg/m ³
Water solubility	ca. 30 g/l (20 °C)
speed of hydrolysis	half-life period: 1 years (25 °C)
Partition coefficient (n-octanol/water)	log Pow: -1,28 Method: (calculated)
Viscosity, dynamic	not applicable
Burning number	BZ 5 - burns out with flames or shower of sparks. Method: VDI 2263
Vapour density	No data available

10. STABILITY AND REACTIVITY

Possibility of hazardous reactions	Must be rendered inert for grinding.
------------------------------------	--------------------------------------

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	6 / 10

Hazardous decomposition products sulphides, flammable smouldering gases

11. TOXICOLOGICAL INFORMATION

Acute oral toxicity	LD50 Rat: > 10000 mg/kg Method: literature No signs of toxicity occurred
Acute inhalation toxicity	LC0 rat(male/female): > 5,25 mg/l / 4 h Method: OECD Test Guideline 403 limit test (maximum concentration attainable in experiments) - No deaths occurred.
Acute dermal toxicity	No test results available.
Skin irritation	Rabbit: 500 mg / 4 h / No skin irritation Method: OECD Test Guideline 404
Eye irritation	Rabbit: 100 mg / No eye irritation Method: OECD Test Guideline 405
Sensitization	Buehler Test guinea pig: not sensitizing to the skin Method: OECD Test Guideline 406
Repeated dose toxicity	Oral Rat Testing period: 9 month NOAEL: 700 mg/kg Method: literature Reversible effects during the application period on liver, spleen, pancreas,
Gentoxicity in vitro	Microorganisms, cell cultures none mutagenic / genotoxic effects Method: literature Ames test Salmonella typhimurium negative Method: OECD TG 471
Carcinogenicity	No test results available.
Toxicity to reproduction	1 generation pharyngeal probe Rat: in maternally non-toxic doses NOEL (No Observed Effect Level) of parents: 300 mg/kg NOEL F1: 300 mg/kg Method: OECD Test Guideline 415
Human experience	gastro-intestinal symptoms: nausea, vomiting Side-effects were observed in the event of higher dosage (10 g)

12. ECOLOGICAL INFORMATION

Elimination information (persistence and degradability)

Biodegradability	aerobic DOC (Dissolved Organic Carbon)
------------------	--

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	7 / 10

inoculum: Activated sludge
Concentration: 17 mg/l
Exposure time: 28 d
Result: 97 % Readily biodegradable.
Method: OECD TG 301 A
DOC decay

Behaviour in environmental compartments

Bioaccumulation low
log Pow: see chapter 9

Ecotoxicity effects

Toxicity to fish LC50 (Brachydanio rerio): > 3200 mg/l / 96 h
Method: OECD 203

NOEC (Brachydanio rerio): 3200 mg/l / 96 h
Method: OECD 203

Toxicity to daphnia EC50 static test Daphnia magna: 324 mg/l / 48 h
Method: OECD TG 202
Own test result.

Toxicity to algae EC50 static test Desmodesmus subspicatus: > 1000 mg/l / 72 h
End point: Biomass
Analytical monitoring: yes
Method: OECD TG 201

EC50 static test Desmodesmus subspicatus: > 1000 mg/l / 72 h
End point: growth rate
Analytical monitoring: yes
Method: OECD TG 201

Toxicity to bacteria EC 10 Pseudomonas putida: 2000 mg/l / 18 h
Method: UBA method

Further information on ecology

Biochemical Oxygen Demand 480 mg/g
(BOD) Concentration: (BOD5)

13. DISPOSAL CONSIDERATIONS

Product

Disposal according to local authority regulations.
Offer rinsed packaging material to local recycling facilities.

Waste Key Number

The waste key number must be determined as per the European Waste Types List (decision on EU Waste Types List 2000/532/EC) in cooperation with the disposal firm / producing firm / official authority.
No waste key number as per the European Waste Types List can be assigned to this product, since such classification is based on the (as yet undetermined) use to which the product is put by the consumer.

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	8 / 10

14. TRANSPORT INFORMATION

Transport/further information

Not dangerous according to transport regulations.

15. REGULATORY INFORMATION

registration

Europe (EINECS/ELINCS)	listed/registered
USA (TSCA)	listed/registered
Canada (DSL)	listed/registered
Australia (AICS)	listed/registered
Japan (MITI)	listed/registered
Philippines (PICCS)	listed/registered
China	listed/registered
Switzerland	not listed/registered

National legislation

16. OTHER INFORMATION

Risk phrase (R phrase) texts

Texts of the H-phrases

Further information

Changes since the last version are highlighted in the margin. This version replaces all previous versions.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	9 / 10

Legend

ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ASTM	American Society for Testing and Materials
ATP	Adaptation to Technical Progress
BCF	Bioconcentration Factor
BetrSichV	German Ordinance on Industrial Safety and Health
c. c.	closed cup (geschlossenes Gefäß)
CAS	Chemical Abstract Services
CESIO	European Committee of Organic Surfactants and their Intermediates
ChemG	German Chemicals Act
CMR	Carcinogenic-Mutagenic-toxic for Reproduction
DIN	German Institute for Standardization
DNEL	Derived No Effect Level
EINECS	European Inventory of Existing Commercial Chemical Substances
GefStoffV	German Ordinance on Hazardous Substances
GGVSEB	German ordinance for road, rail and inland waterway transportation of dangerous goods
GGVSee	German ordinance for sea transportation of dangerous goods
GLP	Good Laboratory Practice.
GMO	Genetic Modified Organism
IATA DGR	International Air Transport Association – Dangerous Goods Regulations
ICAO-TI	International Civil Aviation Organisation - Technical Instructions
IMDG Code	International Maritime Dangerous Goods Code
ISO	International Organization For Standardization
LOAEL	Lowest Observed Adverse Effect Level
LOEL	Lowest Observed Effect Level
NOAEL	No Observed Adverse Effect Level
NOEC	No Observed Effect Concentration
NOEL	No Observed Effect Level
o. c.	open cup (offenes Gefäß)
OECD	Organisation for Economic Cooperation and Development
OEL	Occupational Exposure Limit
PBT	Persistent, Bioaccumulative, Toxic
PEC	Predicted Effect Concentration
PNEC	Predicted No Effect Concentration
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
TA	Technical Instructions (German Ordinance)
TPR	Third Party Representative (Art. 4)
TRGS	Technical Rules for Hazardous Substances (German Regulations)
VCI	German "Verband der Chemischen Industrie e. V."
vPvB	Very Persistent, Very Bioaccumulative
VOC	Volatile Organic Compounds
VwVwS	German Administrative Regulation on the Classification of Substances Hazardous to Waters into Water Hazard Classes
WGK	German Water Hazard Class
WHO	World Health Organization

SAFETY DATA SHEET (EC 1907/2006)

MetAMINO®

DL-Methionine, Feed Grade 99%



Material no.		Version	5.5 / REG_EU
Specification	101612	Revision date	22.11.2011
VA-Nr		Print Date	28.11.2011
		Page	10 / 10

ORIGINAL CONTROLLED